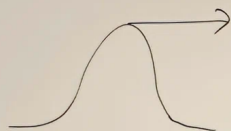


RECALLING THE PROJECT:

① FREE GAUSSIAN WAVEPACKETS



$$\psi(0, x) = \left(\frac{2a}{\pi}\right)^{1/4} e^{-ax^2} e^{+i\frac{p}{\hbar}x}$$

$$\frac{1}{2\pi} \int_{-L}^{+L} d\hbar e^{+i\hbar x} = \delta_L\left(\frac{x}{L}\right)$$

TEST FUNCTION

$$f(x) = \int_{-\infty}^{+\infty} d\hbar e^{-i\hbar x} f\left(\frac{\hbar}{L}\right) \delta_L\left(\frac{x}{L}\right) = f(0) \quad L \rightarrow \infty$$

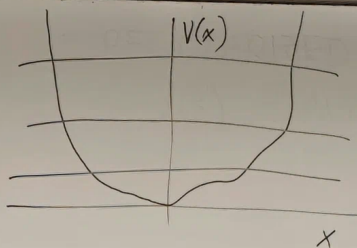


LAST TIME:

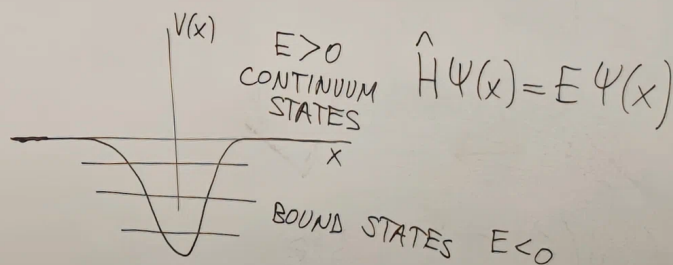
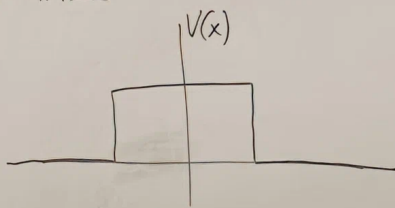
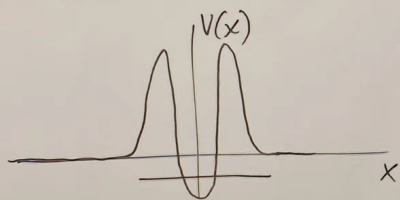
- FREE PARTICLE $\hat{H} = \frac{\hat{p}^2}{2m}$

- BOUND POTENTIALS $\hat{H} = \frac{\hat{p}^2}{2m} + V(x)$

ORTHONORMALITY & CLOSURE



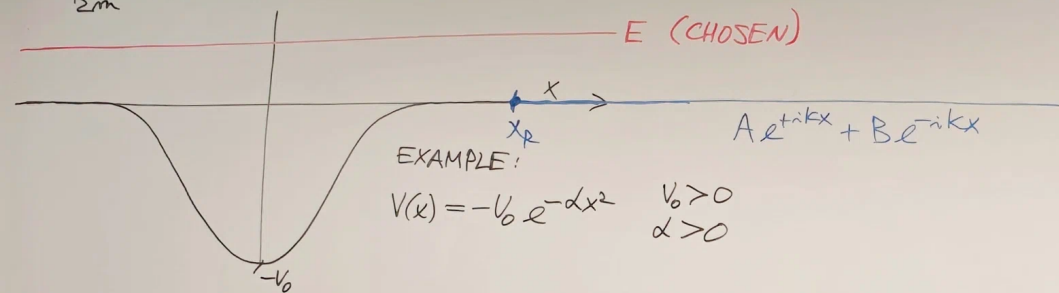
• SHORT RANGED POTENTIALS



TISE

$$-\frac{\hbar^2}{2m} \partial_{xx} \psi(x) + V(x) \psi(x) = E \psi(x)$$

CONTINUUM STATES = SOLUTIONS FOR $E > 0$ OF TISE



EXAMPLE:
 $V(x) = -V_0 e^{-\alpha x^2}$
 $V_0 > 0$
 $\alpha > 0$

LET US CHOOSE SOME FIXED

A, B

WE KNOW

$$\psi(x_R) = A e^{+ikx_R} + B e^{-ikx_R}$$

$$\psi'(x_R) = A i k e^{+ikx_R} - B i k e^{-ikx_R}$$

LET US LOOK FIRST AT THE REGION OF LARGE x :

$x \in (x_R, \infty)$ WHERE x_R IS LARGE ENOUGH SUCH THAT $V(x) \approx 0$ $x > x_R$

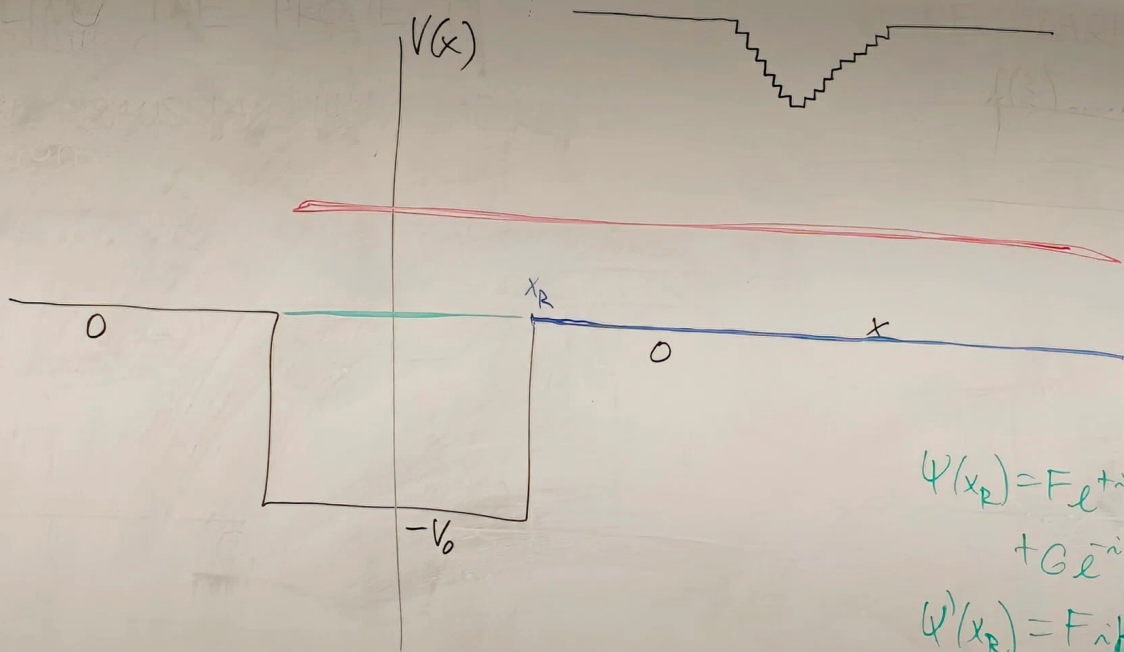
$$-\frac{\hbar^2}{2m} \partial_{xx} \psi(x) = E \psi(x) \quad \text{LIKE WE HAD BEFORE (FREE PARTICLE)}$$

GENERAL SOLUTION:

$$\psi(x) = A e^{+ikx} + B e^{-ikx}$$

$$\hbar k = \sqrt{2mE}$$

↑
ARBITRARY



$$\psi(x_R) = F e^{+ikx_R} + G e^{-ikx_R}$$

$$\hbar k = \sqrt{2m(E+V_0)}$$

$$\psi'(x_R) = F i k e^{+ikx_R} - G i k e^{-ikx_R}$$

$$\psi(x) = F e^{+ikx} + G e^{-ikx}$$

GENERAL SOLUTION

$$-\frac{\hbar^2}{2m} \partial_{xx} \psi(x) - V_0 \psi(x) = E \psi(x)$$

$$-\frac{\hbar^2}{2m} \partial_{xx} \psi(x) = (E + V_0) \psi(x)$$

> 0

SOLUTION OF AN INITIAL VALUE PROBLEM

$$-\frac{\hbar^2}{2m} \partial_{xx} \psi(x) + V(x) \psi(x) = E \psi(x)$$

INITIAL CONDITIONS $\psi(x_0) = \psi_0$
 $\psi'(x_0) = \psi'_0$ GIVEN

$$\psi'(x) \doteq \frac{\psi(x_n) - \psi(x_{n-1})}{\Delta x}$$

$$\psi''(x) \doteq \frac{\psi(x_{n+1}) - 2\psi(x_n) + \psi(x_{n-1}))}{(\Delta x)^2}$$

INSTEAD OF ODE WE HAVE A DIFFERENCE EQUATION

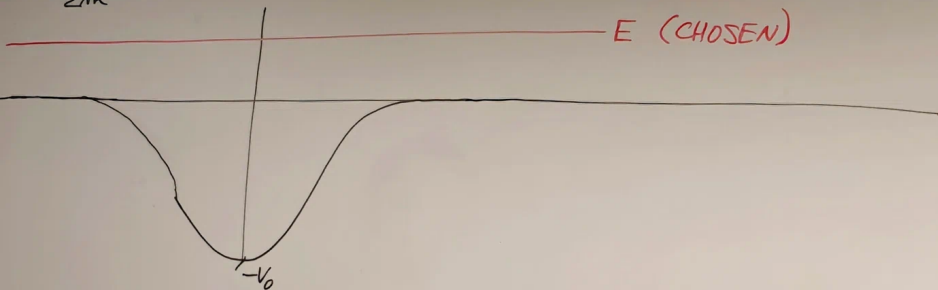
$$-\frac{\hbar^2}{2m} \frac{\psi(x_{n+1}) - 2\psi(x_n) + \psi(x_{n-1}))}{(\Delta x)^2} + V(x_n) \psi(x_n) = E \psi(x_n)$$

$$-\frac{\hbar^2}{2m} \frac{\psi(x_1) - 2\psi(x_0) + \psi(x_{-1}))}{(\Delta x)^2} + V(x_0) \psi(x_0) = E \psi(x_0)$$

TISE

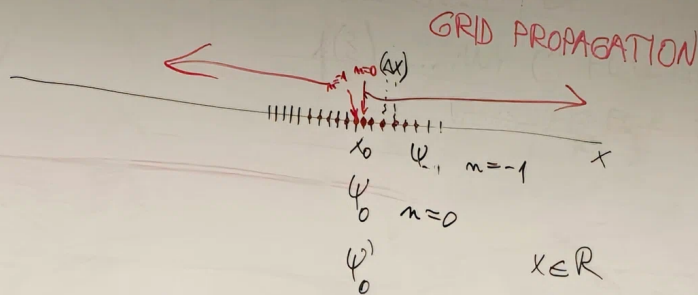
$$-\frac{\hbar^2}{2m} \partial_{xx} \psi(x) + V(x) \psi(x) = E \psi(x)$$

E (CHOSEN)



$$\psi(x \rightarrow -\infty) = C e^{+ikx}$$

UNIQUELY DETERMINED (KNOWN) $\rightarrow D e^{-ikx}$



$m = +1$

$$\psi(x_{n+1}) = \psi(x_n) + \psi'(x_n) \Delta x + \frac{1}{2} \psi''(x_n) (\Delta x)^2 + \dots$$

$$\psi(x_{n+1}) + \psi(x_{n-1}) = 2\psi(x_n) + \psi''(x_n) (\Delta x)^2 + \dots$$

CONTINUUM STATES = SOLUTIONS FOR $E > 0$ OF TISE

LOOKING FOR A PARTICULAR SOLUTION

$\tilde{\psi}(x)$ SATISFYING $A=1$
 $B=0$

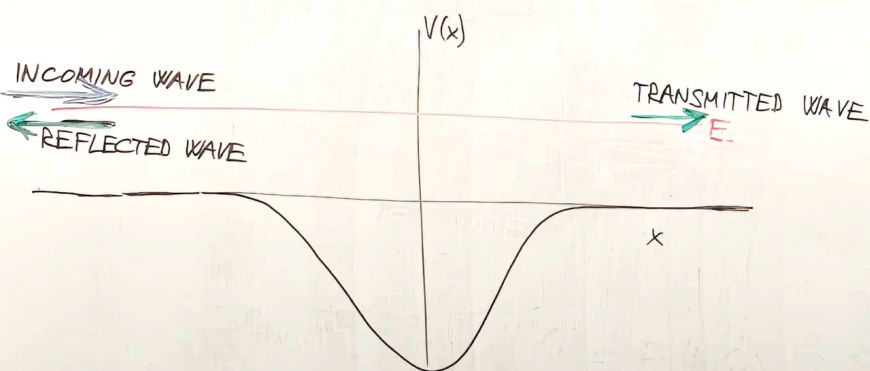
$$\tilde{\psi}(x \rightarrow \infty) = e^{+ikx}$$

FINITE DIFFERENCE METHOD: GET $\tilde{\psi}(x)$ ALONG THE WHOLE REAL AXIS

$$-\frac{\hbar^2}{2m} \partial_{xx} \tilde{\psi}(x) + V(x) \tilde{\psi}(x) = E \tilde{\psi}(x)$$

$$\tilde{\psi}(x \rightarrow +\infty) = e^{+ikx}$$

$$\tilde{\psi}(x \rightarrow -\infty) = C e^{+ikx} + D e^{-ikx}$$



PROBABILITY

$$|T(p)|^2 + |R(p)|^2 = 1 \quad \text{CONSERVATION (*)}$$

$$|T(p)|^2 = \text{TRANSMISSION PROBABILITY}$$

$$|R(p)|^2 = \text{REFLECTION PROBABILITY}$$

$$-\frac{\hbar^2}{2m} \partial_{xx} \psi(x) + V(x) \psi(x) = E \psi(x)$$

SCATTERING BOUNDARY CONDITIONS

$$\psi(x \rightarrow +\infty) = T(p) \frac{1}{\sqrt{2\pi\hbar}} e^{+\frac{i}{\hbar} p x}$$

$$\psi(x \rightarrow -\infty) = \frac{1}{\sqrt{2\pi\hbar}} e^{+\frac{i}{\hbar} p x} + R(p) \frac{1}{\sqrt{2\pi\hbar}} e^{-\frac{i}{\hbar} p x}$$

$$\text{WHERE } p = \sqrt{2mE} > 0, \quad \frac{p^2}{2m} = E.$$

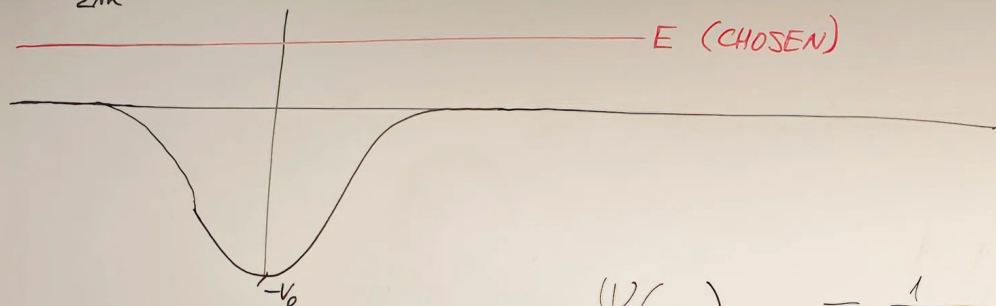
$$T(p) = \text{TRANSMISSION COEFFICIENT} \in \mathbb{C}$$

$$R(p) = \text{REFLECTION COEFFICIENT} \in \mathbb{C}$$

TISE

$$-\frac{\hbar^2}{2m} \partial_{xx} \psi(x) + V(x) \psi(x) = E \psi(x)$$

CONTINUUM STATES = SOLUTIONS FOR $E > 0$ OF TISE



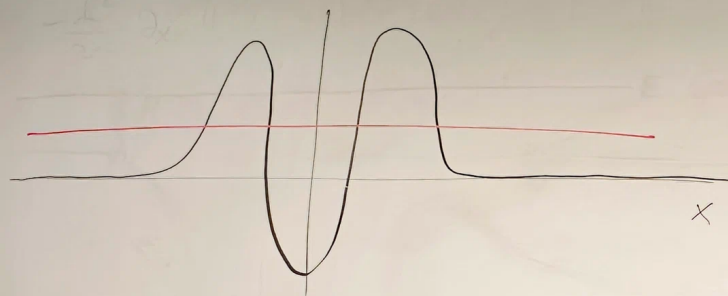
$$\psi(x) = \frac{1}{\sqrt{2\pi\hbar}} \frac{1}{c} \tilde{\psi}(x)$$

$$\frac{D}{c} = R(p)$$

$$\frac{1}{c} = T(p)$$

$$\frac{1}{\sqrt{2\pi\hbar}} \frac{1}{c} \tilde{\psi}(x \rightarrow -\infty) = \frac{1}{\sqrt{2\pi\hbar}} e^{+\frac{i}{\hbar} p x} + \frac{D}{c} e^{-\frac{i}{\hbar} p x} \frac{1}{\sqrt{2\pi\hbar}} = \psi(x \rightarrow -\infty)$$

THIRD PART OF THE PROJECT:



$$V(x) = (0.5x^2 - 0.8)e^{-0.1x^2}$$

$$0 < E < 3$$

CALCULATE $T(p)$, $R(p)$ FOR EACH E

VERIFY (*)

RESONANCE
TUNNELING PHENOMENON

RESONANCE
TUNNELING
DIODES

